Prithvi Narayan Campus

Tribhuvan University

Faculty of Humanities and Social Sciences

Project Report

on

"RaktaSewa"

(A system connecting blood donors and seekers)

Submitted to:

Faculty of Humanities and Social Sciences

BCA program

Prithvi Naryan Campus, Pokhara

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by:

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Tribhuvan University

Faculty of Humanities and Social Sciences

Prithvi Narayan Campus

Supervisor's recommendation

The supervisor wholeheartedly endorses the project titled "RaktaSewa," exclusively developed by **Deep Darshan Thapa and Pratik Pangeni**, acknowledging its exceptional execution and successful fulfillment of the requirements for the Bachelor of Computer Application degree. This project is highly recommended for final evaluation.

Date: 2026/04/04

…………………………..

Gyaneshwor Dhungana

Signature of supervisor

Tribhuvan University

Faculty of Humanities and Social Sciences

Prithvi Narayan Campus

(Date: 2026/04/04)

LETTER OF APPROVAL

This document certifies that **Deep Darshan Thapa** and **Pratik Pangeni's** project, "RaktaSewa" has undergone a comprehensive evaluation to fulfill the requirements for the degree of Bachelor in Computer Application. In our expert opinion, this project demonstrates satisfactory scope and exceptional quality, meeting the standards expected for the degree.

……………………. Gyaneshwor Dungana **Supervisor**	……………………. Bipin Bahadur Adhikari **HOD/Coordinator**
…………………….. **Internal Examiner**	……………………. **External Examiner**

ACKNOWLEDGEMENT

We would like to extend our sincere gratitude to the authorities at Tribhuvan University for providing us with the invaluable opportunity to work on the project titled ‘RaktaSewa'. This project has offered us a platform to apply and explore the practical implications of the theoretical knowledge and expertise we have acquired during our academic journey.

We are deeply appreciative of the unwavering support and guidance extended to us by the Prithvi Narayan Campus throughout this endeavor. We wish to express our heartfelt thanks to our supervisor, **Gyaneshwor Dhungana**, whose scholarly mentorship and valuable suggestions have been instrumental in navigating the complexities of this research. Without his continuous encouragement and belief in the project, this undertaking would have been immensely challenging.

Undoubtedly, this project has not only allowed us to delve deeper into the subject matter but has also significantly enhanced our practical acumen and skills. We are indebted to our friends and teachers who generously shared their insights and perspectives, which have greatly contributed to the improvement and enrichment of this project work.

Furthermore, we extend our deepest appreciation to all individuals who have directly or indirectly contributed to the successful completion of this project. Their guidance and encouragement have been a source of motivation throughout this journey.

With heartfelt thanks,

Deep Darshan Thapa

Pratik Pangeni

Abstract

RaktaSewa is an **Android-based mobile application** designed to provide a faster, smarter, and more reliable way to connect blood donors with recipients. In Nepal, the absence of a centralized, real-time donor–recipient matching system often results in delays that can be life-threatening. This project aims to address this problem by developing a mobile platform that integrates **Firebase** for user authentication, real-time data storage, and instant communication between donors and seekers. The system will be using three core algorithms to enhance matching efficiency. The **Donor Matching Algorithm** filters donors based on blood group compatibility, location proximity, and availability status. The **Emergency Priority Algorithm** ensures urgent requests receive immediate attention by notifying the nearest eligible donors within a specific radius. The **Donor Availability Prediction Algorithm** evaluates eligibility based on the recommended 90-day donation interval and historical records. The project aims to reduce search time, improve donor engagement, and contribute to a more organized and effective blood donation network.

Keywords: RaktaSewa, Android-based mobile application, Firebase, Donor Matching Algorithm, Emergency Priority Algorithm, Donor Availability Prediction Algorithm

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List of Abbreviations

GPS	Global Positioning System
Max	maximum
App	application
FCM	Firebase Cloud Messaging
Auth	authentication
UI	user interface
Lat	latitude
Lng	longitude
Db	database

CHAPTER 1: INTRODUCTION

1.1 Introduction

Blood plays a vital role in saving lives during medical emergencies, major surgeries, accidents, childbirth complications, and chronic illnesses. Despite this, timely access to blood in Nepal continues to be a significant challenge. Many regions across the country frequently experience shortages, and families are often required to arrange blood on their own due to insufficient supply in hospitals and blood banks. For example, hospitals in Banke and Nepalgunj have reported repeated shortages where relatives must independently search for donors when the blood bank cannot meet demand [1]. A similar situation has been reported in Jhapa, where the Red Cross noted that blood supply levels often drop to a critically low level, sometimes lasting only a few days [2].

These difficulties underline the absence of a centralized, up-to-date, and easily accessible donor information system in Nepal. Current blood search practices primarily depend on personal contacts, manual phone calls, and social media requests, which can consume crucial time during emergencies. Additional issues such as irregular donor participation, outdated donor details, and limited access in rural locations further complicate the emergency response process [3].

To help address these challenges, the RaktaSewa system has been developed as an Android-based application that offers real-time data management, location-aware matching, and automated donor–seeker coordination. By enabling quick access to available donors and streamlining communication, the system aims to support faster, more organized, and reliable blood search across Nepal.

1.2 Problem statement

Nepal continues to face recurring challenges in ensuring timely access to blood during emergencies. Many hospitals and blood banks operate with limited and inconsistent stock levels, compelling families to search for donors on their own when supplies run out. Reports from multiple districts such as Banke, Nepalgunj and Jhapa indicate frequent shortages that hinder emergency [2] [1]. Additionally, there is no unified and continuously updated digital donor database, and most donor information remains scattered, outdated, or dependent on personal contacts and social media postings [4] 1 Therefore, there is a need for a digital solution that can intelligently match donors with recipients, provide updated availability status, and support rapid communication during emergencies.

1.3 Objectives

The project was carried out with the the following objectives:

- To design a mobile application that allows users to register as donors or seekers.
- To develop a donor matching algorithm based on blood group compatibility, location, and availability
- To reduce the time taken to find compatible donors during critical situation.

1.4 Scope and limitation

Scopes include:

Platform Development: Development of a mobile-based platform that connects blood donors and seekers through real-time data updates and an intuitive user interface.

Donor and Seeker Management:

Users can register as donors or seekers, maintain profiles, and update their availability status.

Blood Request Management:

Seekers can create blood requests by entering patient details, location, and urgency levels. Requests are stored and processed in real-time.

User Authentication:

The application includes secure user login and verification to protect user data and ensure trusted interactions.

RaktaSewa has a few limitations. Some of the limitations of this application are:

Limited Donor Base: In the early stages, the number of seekers and donors on the platform may be limited, affecting the variety and availability of blood for needly ones.

User Adoption: The system's effectiveness relies heavily on users actively installing the app, updating availability, and responding to notifications.

Medical eligibility: The eligibility of the donor completely depends upon the form filled.

Location variation: Location accuracy may vary depending on device GPS and network condition and currently donor and seeker residing on.

Overall, the project aims to create a viable and user-friendly platform for searching for blood available at emergency, but its success will depend on addressing these limitations and continuously adapting to meet seekers demands and condition.

1.5 Development methodology

This project follows the Prototyping Model for system development. The prototyping model is a software development methodology in which an initial version or prototype of the system is developed to understand the requirements more clearly and gather feedback before the final implementation. This methodology helps in identifying user needs, reducing development risks, and improving system usability. This methodology was chosen because it allows continuous improvement of the system through feedback and ensures that the developed system meets user requirements effectively.

1.6 Report organization

Chapter 1: Topic "Introduction" contains the introduction to the project. It explains the project's objectives, scope, and limitations, as well as provides a brief description and summary. It also explains why we're working on this project.

Chapter 2: Topic "Background Study and Literature Review" presents a critical evaluation of the context of our system-critical analysis of existing literature. It includes a description of our perspective of the previous system, as well as what our project wants to accomplish, as well as a comparison of the traditional and our planned system.

Chapter 3: Topic, "System Analysis and Design," includes a data flow diagram, modules, and architecture, as well as contains the requirement of the system, its hardware requirement, and the software required to run our system. It also relates to shaping organizations, improving performance, and achieving objectives for profitability and growth.

Chapter 4: Topic "Implementation and Testing" includes the process of testing implementations of technical specifications. It consists of all the functions and how the functions are implemented and what functions are used to implement them.

Chapter 5: Topic, "Conclusion and Future Recommendations," includes how well we have achieved our original aim and objectives, what the limitations and scopes of our system are, what we should do differently next time, and the consequences for research funding and practice.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background study

As part of the background study for the RaktaSewa project, research and analysis were conducted to understand the challenges of the blood donation system in Nepal. Through online research and discussions with potential users, it was found that there is no centralized and real-time platform to connect blood donors with recipients. People often depend on social media or personal contacts, which leads to delays that can be life-threatening in emergency situations.

Existing systems were analyzed to identify their strengths and limitations. It was observed that many platforms lack real-time updates, efficient donor filtering, and proper communication features. Users expressed the need for a reliable system that ensures quick response, accurate matching, and secure data handling.

With the increasing use of smartphones and internet access, an Android-based solution was identified as an effective approach. The RaktaSewa application is designed to provide a faster, smarter, and more reliable way to connect donors and recipients. It integrates Firebase for secure user authentication, real-time data storage, and instant communication.

The system uses three core algorithms to improve efficiency. The Donor Matching Algorithm filters donors based on blood group, location, and availability. The Emergency Priority Algorithm prioritizes urgent requests by selecting the nearest suitable donors. The Donor Availability Prediction Algorithm determines donor eligibility based on their last donation date and recommended intervals.

Overall, the project aims to reduce search time, improve donor engagement, and create a more organized and effective blood donation network. The background study played a key role in shaping the system to meet real-world needs and provide a reliable life-saving solution.

2.2 Literature Review

Blood donation and management systems have been a focus of research and development worldwide due to their critical role in healthcare and emergency response. In Nepal, the Nepal Red Cross Society manages blood inventory and donor information primarily through manual processes and offline records. While this system provides essential services, it often faces challenges in ensuring timely availability of blood during emergencies, particularly in remote regions where access to updated donor information is limited. Reports from districts like Banke, Nepalgunj, and Jhapa indicate frequent shortages, compelling families to rely on personal networks or social media to find donors. This demonstrates a clear gap in accessibility and real-time communication in existing systems [1] [2].

Studies on blood donation systems in Nepal and other developing countries highlight several recurring challenges. These include lack of a centralized and continuously updated donor database, delays in communication between donors and seekers, limited access in rural areas RaktaSewa is proposed to address these gaps by providing an android based platform that integrates a centralized Firebase database for real-time donor information storage and retrieval. The system uses algorithms to match donors and seekers based on blood group compatibility, location, and availability reducing the time taken to find compatible donors.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

To ensure effective development, the Prototyping Methodology was adopted. This approach allows the development of an early working version of the application, which is then refined through continuous user feedback. As RaktaSewa involves features that must be tested directly with users (such as donor availability toggles, GPS accuracy, and notification flow), prototyping proved highly suitable. Feedback from trial users helped refine request forms, matching logic, and user interface elements, ensuring the final system closely aligns with real-world needs.

By iteratively analyzing user requirements, validating features through prototypes, and refining the workflow based on feedback, the system analysis phase ensured that RaktaSewa was developed as a practical, efficient, and user-centered blood management solution.

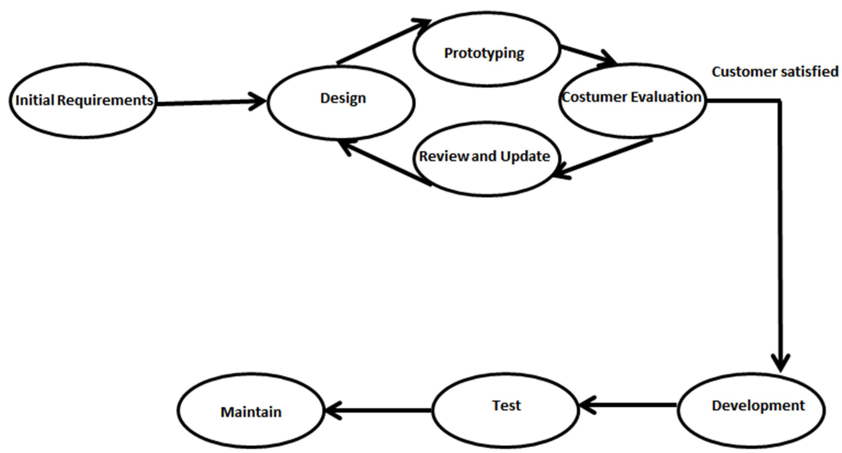


Figure 1 Prototyping Methodology

3.1.1 Requirement Analysis

The functional requirements define the specific features and operations that the RaktaSewa system must provide. These requirements were identified based on user needs, existing challenges in blood search, and system objectives.

Functional Requirements:

User Registration & Authentication

- The system shall allow users to register as either donors or seekers.
- Users shall be able to log in securely using email and password.
- The system shall allow users to update their personal profile (name, phone number, blood type, etc.).

Donor Features

- Donors shall be able to update their availability status (available / unavailable).
- Donors shall be able to record their last blood donation date.
- Donors shall receive real-time notifications when a matching request is created.
- Donors shall be able to view details of a blood request and accept it.

Seeker Features

- Seekers shall be able to create a blood request by providing patient details, blood group, location, and urgency.
- Seekers shall be able to find donors by blood group and proximity.
- Seekers shall be notified when a donor accepts their request.
- Seekers shall be able to mark a request as fulfilled.

Matching & Notification System

- The system shall automatically match donors and seekers based on:
    - Blood group compatibility
    - Location proximity
    - Donor availability
- The system shall send push notifications to eligible donors.
- The system shall log notification history.

Non-functional requirements:

Performance Requirements

- Notifications should be delivered within 2–3 seconds after a request is created.
- The system shall return donor search results in real time (under 1 second for database queries).

Security Requirements

- User data shall be stored securely using Firebase Authentication and Firestore security rules.
- Sensitive data (phone numbers, locations) shall not be accessible without proper authorization.

Usability Requirements

- The application interface shall be simple, intuitive, and easy to navigate.
- Important actions (Create Request, Accept Request, Update Availability) shall require minimal steps.
- The app shall support clear visual indicators such as color-coded statuses and alerts.

System Flow chart

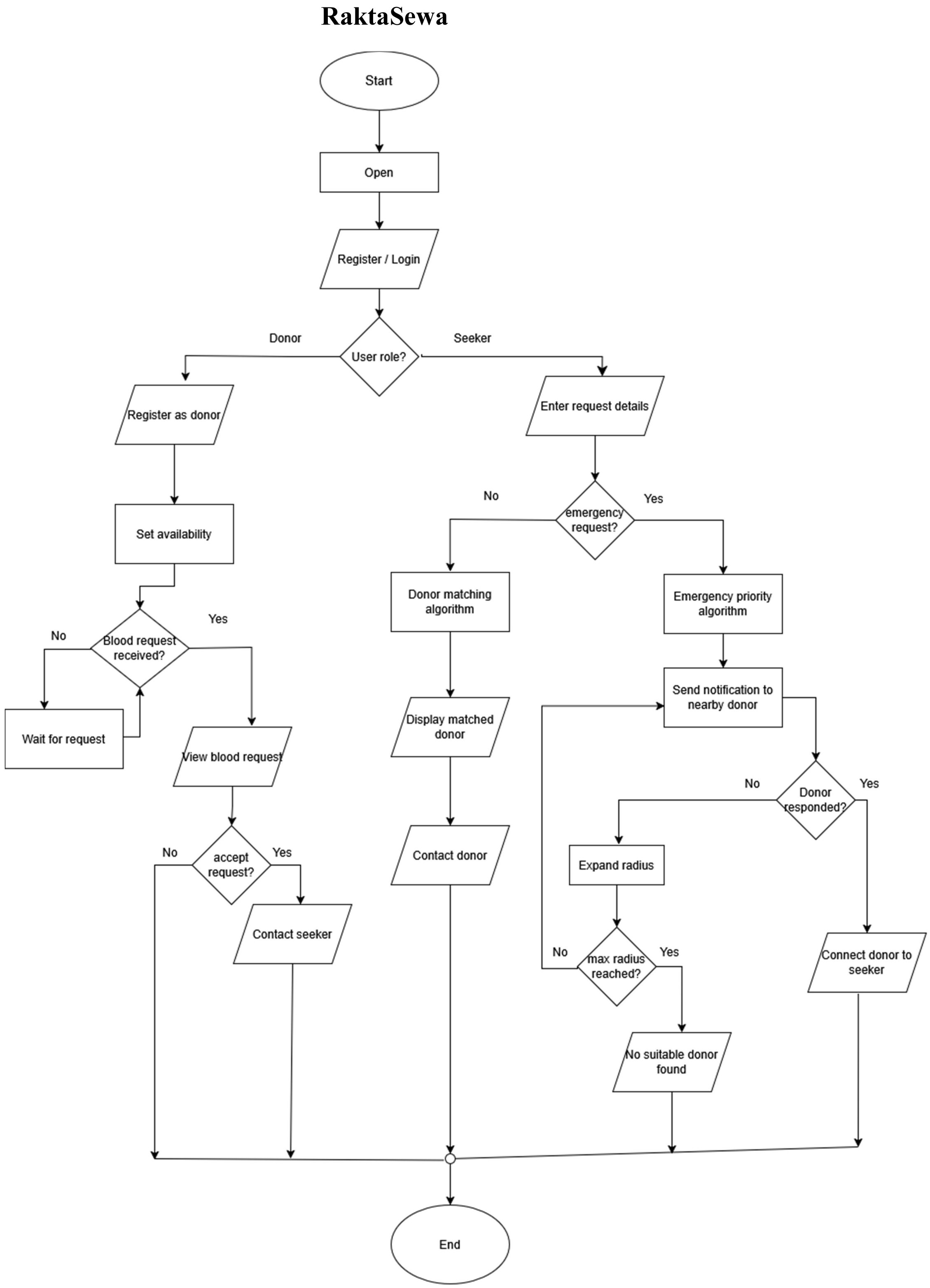


Figure 2 System Flow Chart

Use Case Diagram:

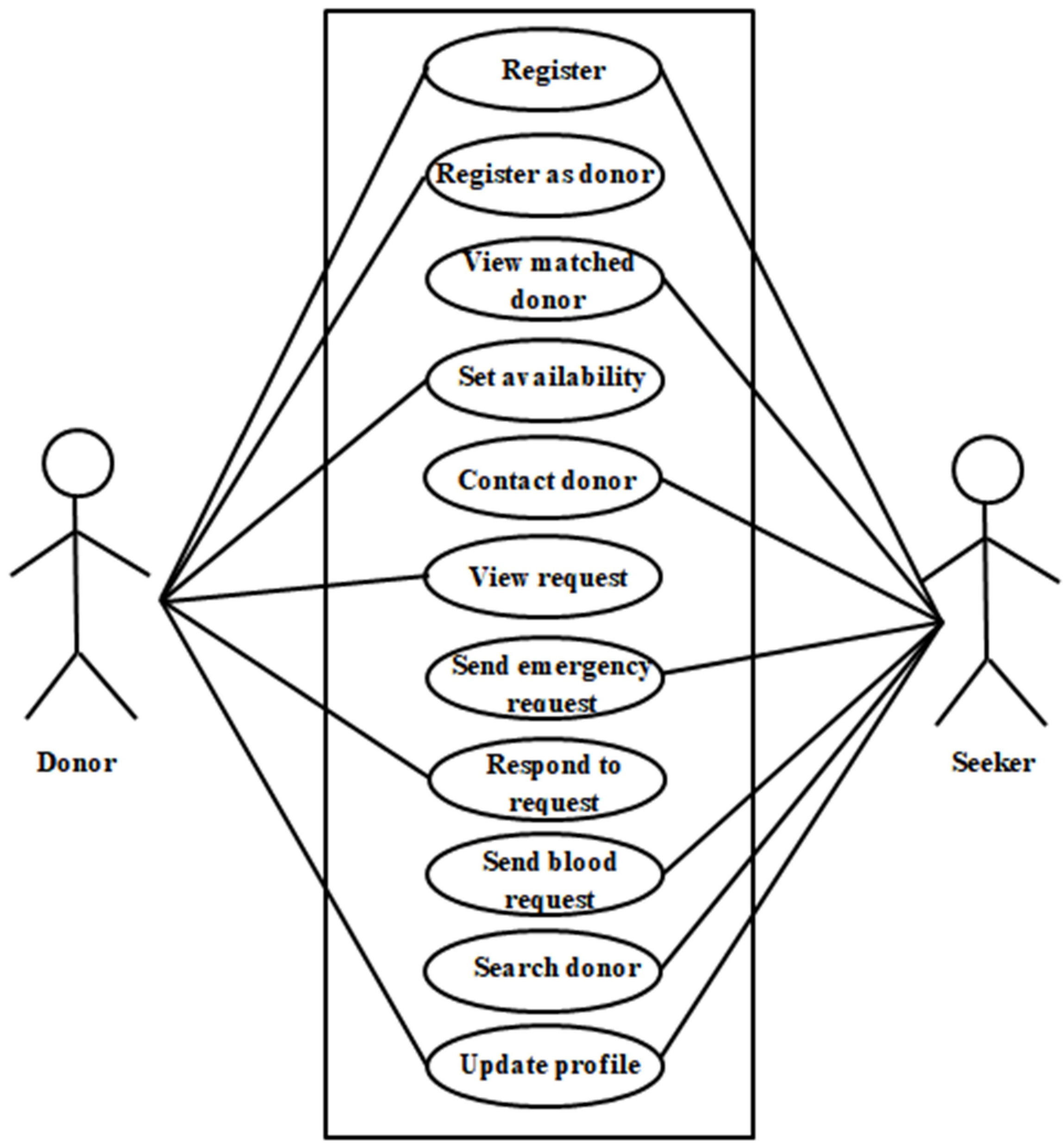


Figure 3 Use case diagram

3.1.2 Feasibility study:

A feasibility study was conducted for the project to see if a system idea is feasible. A feasibility study has been conducted to assess the viability of the RaktaSewa, covering technical, economic, and legal aspects.

- **Operational feasibility:**

  This project is operationally feasible as it is reliable for all types of users with stable internet connectivity. Operational Feasibility focuses on determining if this project has the operational capability and resources to manage and maintain a mobile application platform once it is developed.

- **Economical feasibility:**

  Our system utilizes free software which will not generate any costs also, the application does not spend much more money, so our system is economically feasible.

- **Technical feasibility:**

  Technical feasibility centers on evaluating whether the necessary technology is available and capable of supporting development. After the technical needs have been carefully analyzed, we have the technical skills and resources needed to system. The user interface of the app is clear and simple to use. It is made sure the website works technically.

- **Legal Feasibility:**

  Research has been conducted on legal requirements for operating a RaktaSewa. Steps have been taken to ensure compliance with data privacy laws and regulations.

3.1.3 System Design

The app is designed to be simple and easy to use. Donors can update their status & location and seekers can search for donors, and the system connects them automatically.

3.1.4 GANTT chart

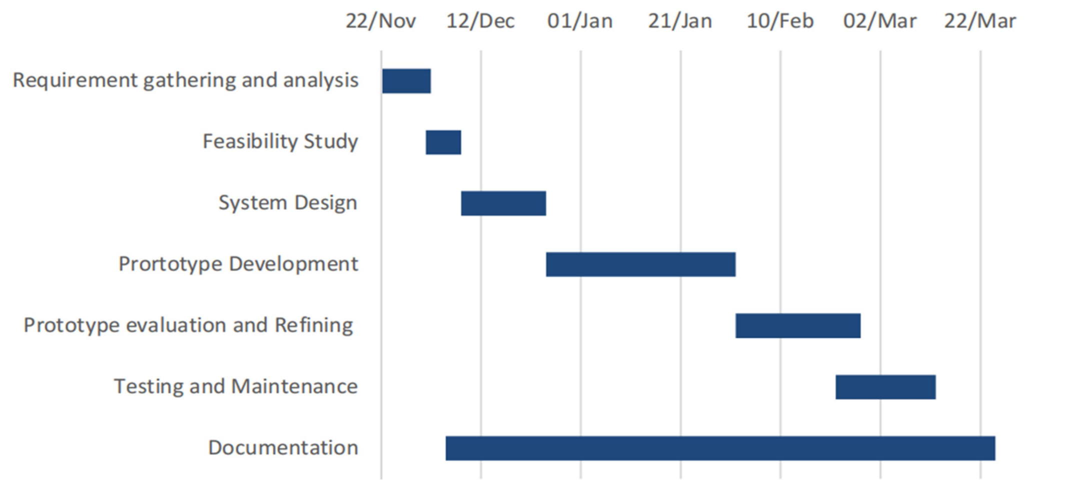


Figure 4 Gantt Chart

3.1.5 E-R Diagram

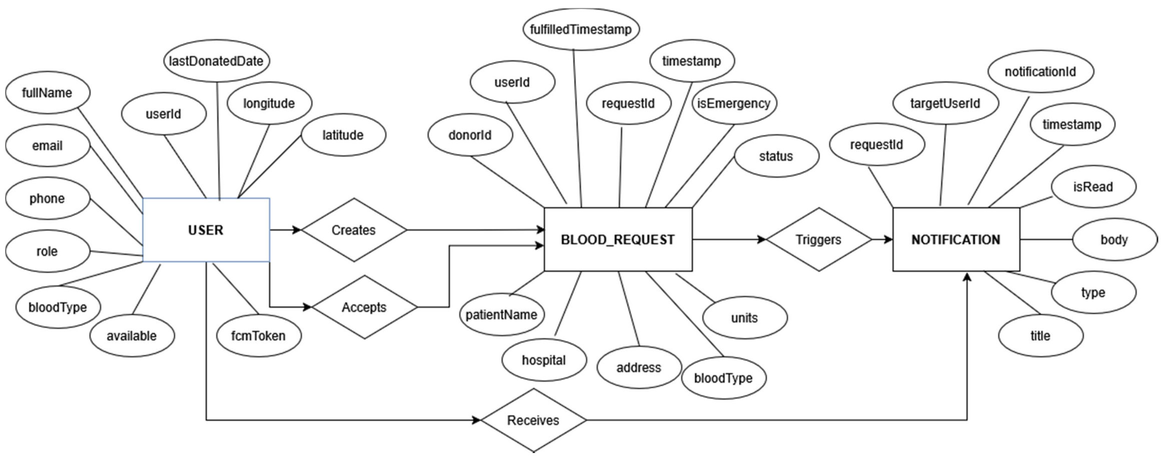


Figure 5 ER Diagram

3.1.6 Process Modeling (DFD)

DFD level 0

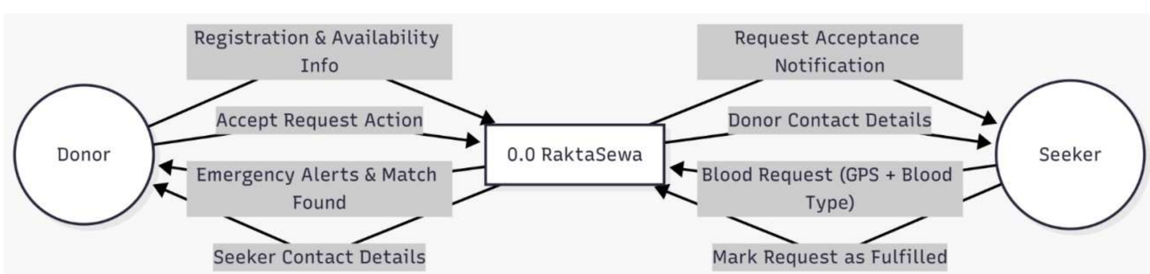


Figure 6 DFD Level 0

DFD Level 1

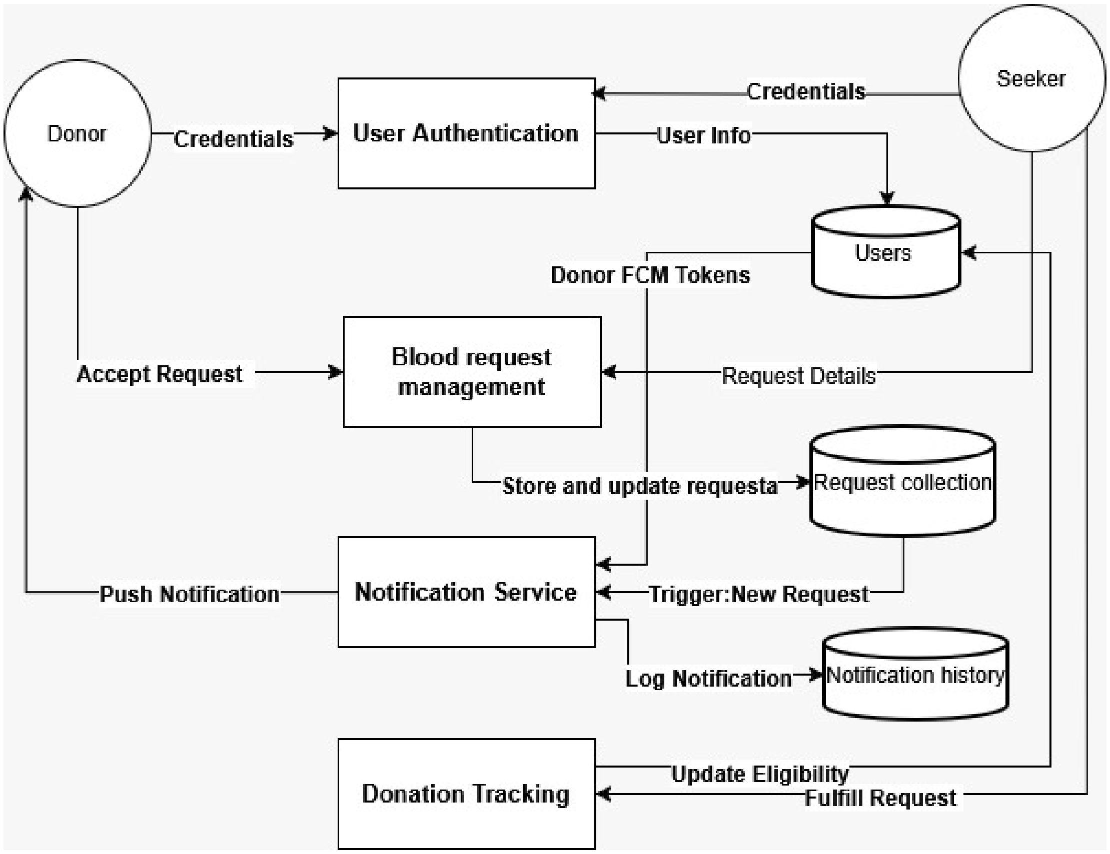


Figure 7 DFD Level 1

3.2 System Design

System design is the process of defining elements of a system like modules, architecture, components, and their interfaces and data for a system based on the specified requirements. System design is the process of defining, developing, and designing systems that satisfy the specific needs and requirements of an organization.

3.2.1 Architectural Design

In architectural diagram for a software project is a visual representation of the system's overall structure and how its components interact with each other. In the case of an online tutor project, the architectural diagram provides a high-level overview of the system and its various components, including the user interface, database, application server, and external systems or services. The diagram helps to illustrate how these components fit together and how data flows through the system. It can also help to identify potential bottlenecks or areas of the system that may require optimization.

3.2.2 Interface Design

Designing the interface for RaktaSewa involves creating user-friendly and intuitive screens for users as well as an admin panel for overseeing the platform's operations. Here's a breakdown of the interface design:

1. **Login Page:**

   Login: The user needs to login firstly. If the user doesn't have logged in yet and is new then the user must register by clicking register here.

   Register As Donor: If the user wants and is eligible to donate blood then the can register as donor.

   Register As Seeker: If the user is seeking for blood then they can request for their blood type filling essential details.

2. **Home Page:**

   Header: Contains the application logo, and user login/register options.

   Footer: contains navigation buttons like home, donors, requests, profile.

   Blood Request Listings: Display a grid of blood requests, each showing a blood details, type, patient details, and location. Users can scroll through these listings.

3. **Donor Profile:**

   Donor Dashboard: After login, users are directed to their dashboard, which includes options for viewing accepted requests, log donation date, donation history, managing requests and logout.

   Accepted Requests: Display a list of blood the donor has listed for donation. Include options to edit and remove.

   Donation History: A section where users can see total of donataion he/she has made.

4. **Seeker Profile:**

   Profile Edit: The seeker can edit their profile details.

Register As Donor: The seeker can also register as donor and donate blood too.

Managing Owns Requests: The seeker can manage their own requests of blood query.

3.3 Algorithm details

This system incorporates multiple algorithms to enhance efficiency, accuracy, and real-time responsiveness. These algorithms are categorized as follows:

<u>1. Geospatial and Proximity Algorithms</u>

The system uses location-based algorithms to identify nearby donors and optimize search performance.

- Haversine-Based Distance Calculation:

The application uses the Location.distanceBetween() method to calculate the distance between donor and seeker coordinates. This helps in determining proximity accurately.

- Incremental Radius Expansion Search:

The search process begins within a 5 km radius and gradually expands by 5 km up to 25 km if no donors are found. If still unsuccessful, a global search is performed. This reduces unnecessary database load while prioritizing nearby donors.

- Geofencing for Emergency Notifications:

A 10 km radius is defined around the hospital location. All donors within this boundary are identified and notified instantly during emergencies.

<u>2. Healthcare Eligibility Algorithm</u>

- 90-Day Deferral Algorithm:

The system calculates the time difference between the current date and the donor’s last donation date. If the difference is less than 90 days, the donor is marked ineligible. Otherwise, the donor is eligible. The system also displays remaining days for eligibility.

3. Search and Prioritization Algorithms

- Emergency-First Sorting:

Requests are sorted based on emergency status (priority) and timestamp (recency), ensuring urgent cases appear first.

- Index Fallback Sorting:

If database sorting fails, the system applies client-side sorting using Collections.sort() to maintain proper order.

4. Data Lifecycle Management Algorithm

- Auto-Expiry Logic:

Completed requests are automatically removed from the dashboard after 24 hours to maintain data relevance.

5. Notification Routing Algorithm

- Conditional Dispatch Logic:

In normal cases, notifications are sent to donors matching the required blood type. In emergency cases, all donors within a 10 km radius are notified.

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

In this chapter, we focus on transforming the technical design of the RaktaSewa system into a functional mobile application. The implementation phase involves developing the frontend and backend, integrating the database, and selecting appropriate technologies to ensure efficiency and reliability.

System Development Phases

The implementation process consists of several key stages. Firstly, system requirements are analyzed and refined to ensure clarity and proper understanding. Secondly, suitable technologies such as Java, Firebase, and Android Studio are selected. Thirdly, the frontend is developed to provide a user-friendly interface, while the backend handles data processing and communication. Finally, a real-time database is integrated to manage user and donor information efficiently.

System Enhancements

Various features are implemented to enhance the system's performance. User authentication ensures secure login and registration of donors and recipients. The donor matching system filters users based on blood group, location, and availability. The emergency priority feature highlights urgent blood requests for faster response. Additionally, donor availability prediction improves reliability by checking eligibility based on donation history.

Operational Guidelines

This section provides instructions for effective system usage. Users can register, search for donors, request blood, and respond to emergency needs. The system ensures smooth interaction between donors and recipients through real-time updates. Proper usage of the application helps reduce delays and improves response time in critical situations.

4.1.1 Tools Used

We use different tools and techniques for the development of the system. We use Java for the frontend and Firebase for the backend development. We used Android studio for building the project.

Java:

Java is the primary programming language used in this project. It is platform-independent, object-oriented, and widely used for developing secure and robust applications.

Firebase:

Firebase is a Backend-as-a-Service (BaaS) platform developed by Google. It is used in this project to handle backend functionalities such as data storage, authentication, and real-time updates without managing a separate server.

Android Studio:

Android Studio is the official Integrated Development Environment (IDE) for Android app development, developed by Google. It is used in this project to design the user interface, write and manage code, and test the application efficiently.

Vercel:

Vercel is used for backend to host notification server for this project.

Git:

Git is a distributed version control system used to track changes in source code and manage project versions efficiently.

GitHub:

GitHub is an online platform used to store, manage, and collaborate on code using Git. It helps in team collaboration and project management.

Draw.io:

Draw.io is an online diagramming tool used to create flowcharts, UML diagrams, and system designs for better visualization of the project structure.

4.2 Unit testing

Unit testing ensures that each module of RaktaSewa works correctly and meets its functional requirements. Key components like user registration, login, donor search, and blood request handling are tested individually. By using different data inputs, errors are identified and fixed early. Critical features, including donor matching and emergency request processing, are carefully validated to ensure accuracy, reliability, and smooth performance before full system integration.

Table 1 Unit Testing cases

Test ID	Component	Test Description	Expected Output	Status
U-01	Auth	Donor Registration	Account created; location required	Pass
U-02	GPS	Fetch Location	Accurate Lat/Lng + address	Pass
U-03	Logic	Blood Type Matching	Only compatible types appear	Pass
U-04	Eligibility	90-Day Cool-down	Donor cannot toggle availability	Pass
U-05	UI	Copy Hospital Address	Address copied to clipboard	Pass
U-06	Profile	Edit Profile	Data updated in Firebase	Pass
U-07	UI	Contact Requester	Dialer opens with number	Pass

Table 2 System testing cases

Test ID	Test Description	Expected Outcome	Status
S-01	Create Emergency Request	Nearby donors receive alert <2s	Pass
S-03	Accept Blood Request	Requester notified; status = Accepted	Pass

S-04	Toggle Availability	Updates in DB + visible to seekers	Pass
S-05	View Donation History	List of fulfilled requests displayed	Pass
S-06	Delete Notifications	Notification removed from DB + UI	Pass

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATION

5.1 Outcome

This project helped us gain practical experience by applying the knowledge learned in previous courses. Implementing different concepts, tools, and techniques in a real-world application made learning more meaningful. Although the development process was challenging, it was a valuable experience. The RaktaSewa application is user-friendly, efficient, and designed with simple yet effective functionalities to meet user requirements and improve the blood donation process.

5.2 Conclusion

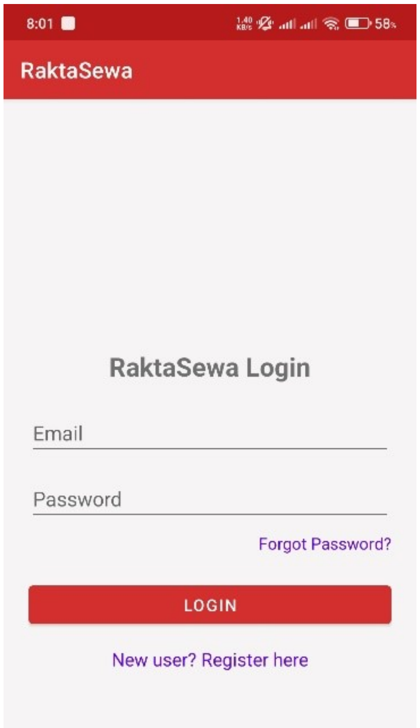
In conclusion, the RaktaSewa project demonstrates significant value for society by improving the process of blood donation and emergency response. The system effectively connects donors and recipients, reducing delays in critical situations. It promotes a sense of community by encouraging people to help one another during emergencies. Overall, the application provides a reliable, efficient, and user-friendly platform that supports timely blood availability and contributes to saving lives.

5.3 Future recommendation

In addition to the current features, the RaktaSewa system can be further improved in several ways:

- Hospital and Blood Bank Integration
- Implementation of Medical Verification for Donors
- Introduce Donor Reward and Engagement System
- Expand to iOS and Web Platforms

Appendices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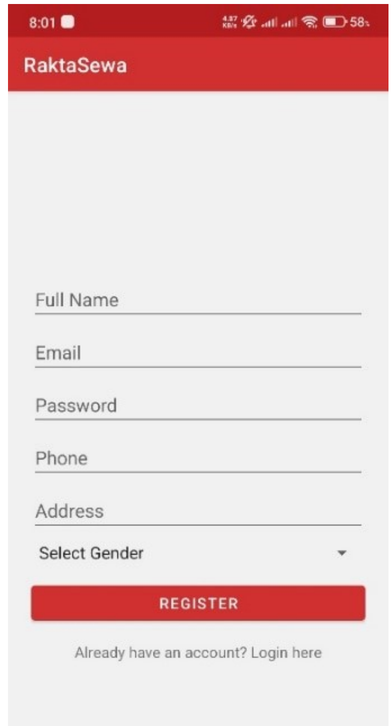
RaktaSewa
RaktaSewa Login
Email
Password
Forgot Password?
LOGIN
New user? Register here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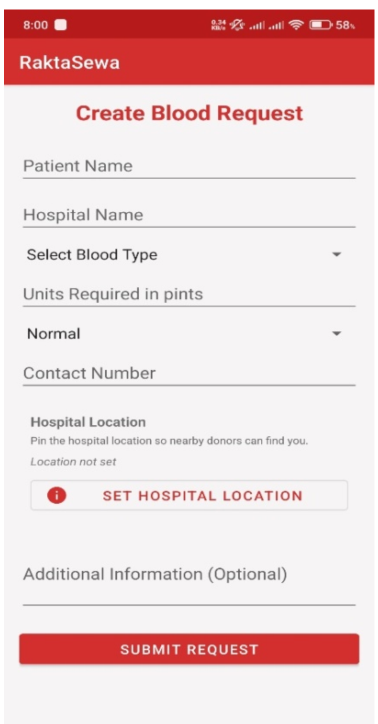
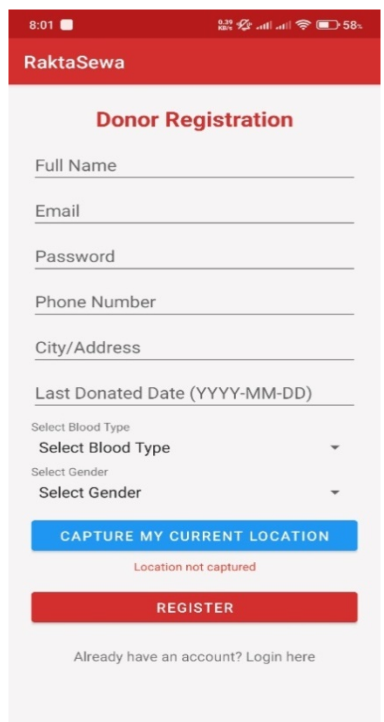
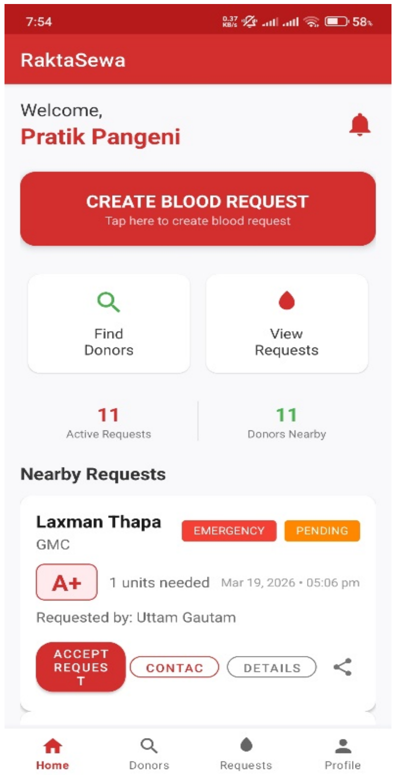
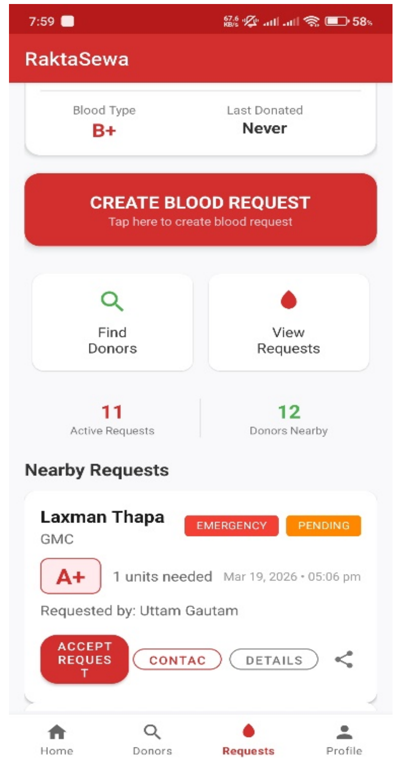
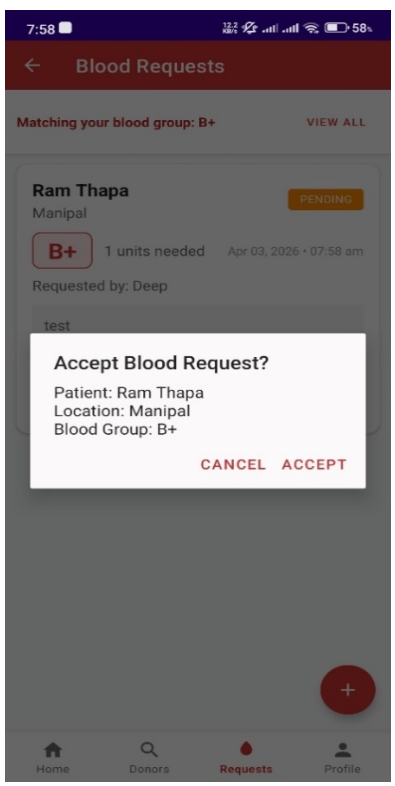
RaktaSewa
Full Name
Email
Password
Phone
Address
Select Gender
REGISTER
Already have an account? Login here

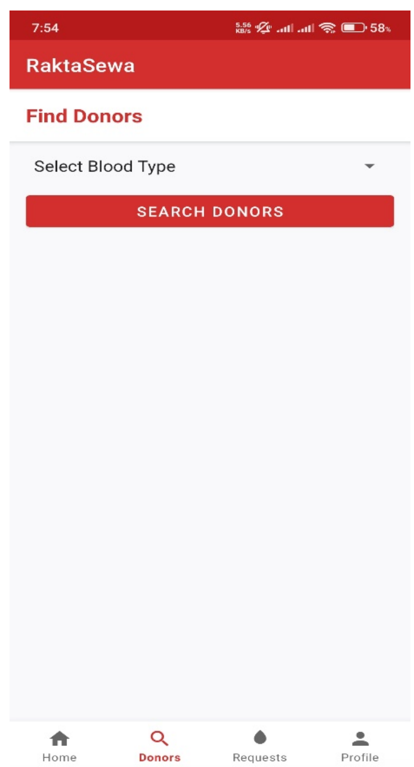

8:00
RaktaSewa
Create Blood Request
Patient Name
Hospital Name
Select Blood Type
Units Required in pints
Normal
Contact Number
Hospital Location
Pin the hospital location so nearby donors can find you.
Location not set
SET HOSPITAL LOCATION
Additional Information (Optional)
SUBMIT REQUEST

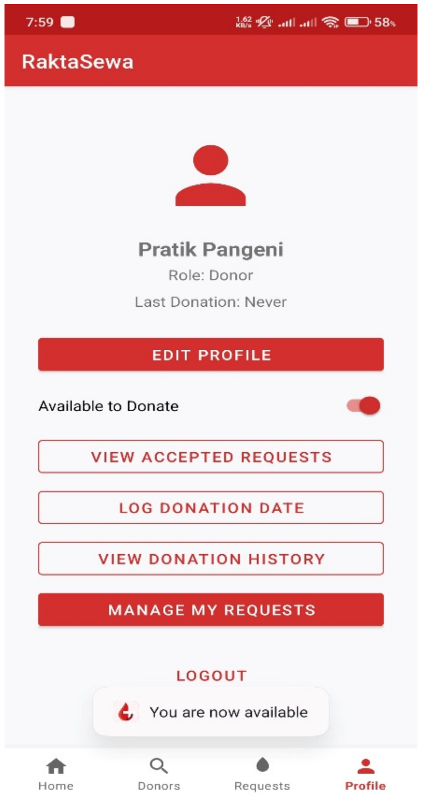

8:01
RaktaSewa
Donor Registration
Full Name
Email
Password
Phone Number
City/Address
Last Donated Date (YYYY-MM-DD)
Select Blood Type
Select Blood Type
Select Gender
Select Gender
CAPTURE MY CURRENT LOCATION
Location not captured
REGISTER
Already have an account? Login here

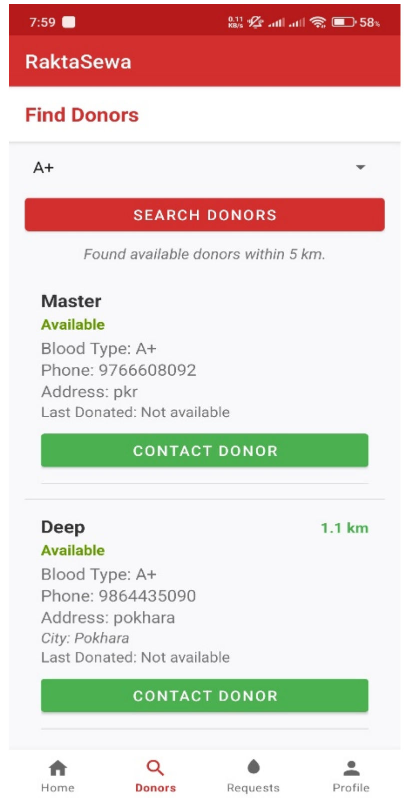

7:54
58%
RaktaSewa
Welcome,
Pratik Pangeni
CREATE BLOOD REQUEST
Tap here to create blood request
Find Donors
View Requests
11
Active Requests
11
Donors Nearby
Nearby Requests
Laxman Thapa
EMERGENCY
PENDING
GMC
A+
1 units needed
Mar 19, 2026 • 05:06 pm
Requested by: Uttam Gautam
ACCEPT REQUEST
CONTAC
DETAILS
Home
Donors
Requests
Profile

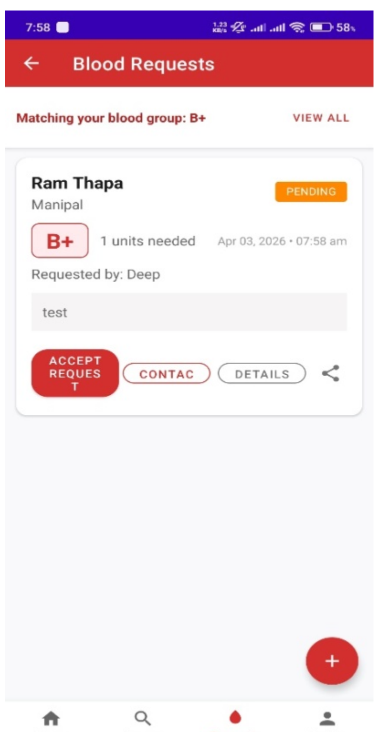

7:59
58%
RaktaSewa
Blood Type
B+
Last Donated
Never
CREATE BLOOD REQUEST
Tap here to create blood request
Find Donors
View Requests
11
Active Requests
12
Donors Nearby
Nearby Requests
Laxman Thapa
EMERGENCY
PENDING
GMC
A+
1 units needed
Mar 19, 2026 • 05:06 pm
Requested by: Uttam Gautam
ACCEPT REQUEST
CONTAC
DETAILS
Home
Donors
Requests
Profile

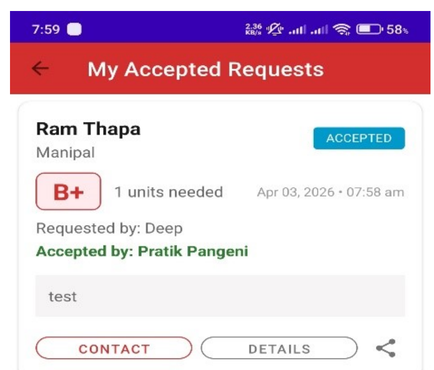

7:58
58%
Blood Requests
Matching your blood group: B+
VIEW ALL
Ram Thapa
PENDING
Manipal
B+
1 units needed
Apr 03, 2026 • 07:58 am
Requested by: Deep
test
Accept Blood Request?
Patient: Ram Thapa
Location: Manipal
Blood Group: B+
CANCEL
ACCEPT
Home
Donors
Requests
Profile

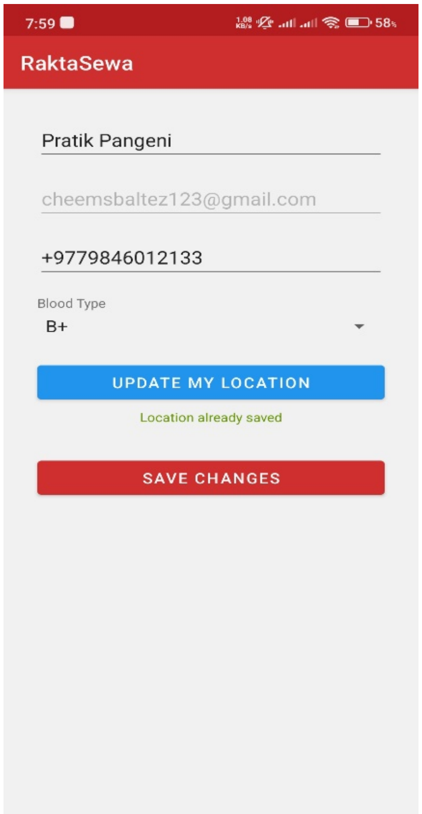

7:54
58%
RaktaSewa
Find Donors
Select Blood Type
SEARCH DONORS
Home
Donors
Requests
Profile

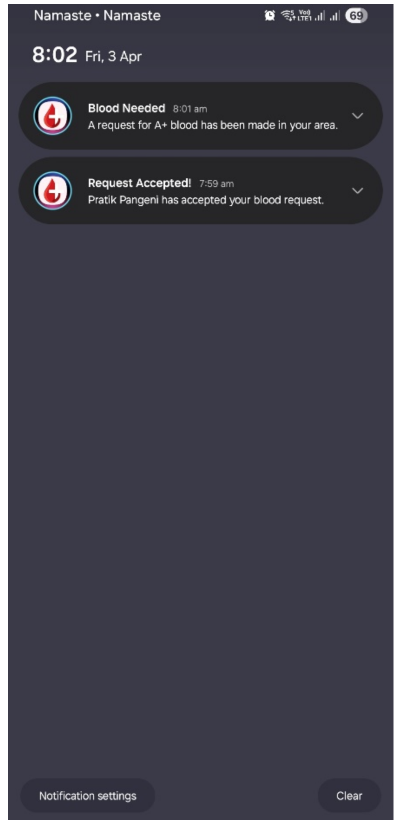

7:59
58%
RaktaSewa
Pratik Pangeni
Role: Donor
Last Donation: Never
EDIT PROFILE
Available to Donate
VIEW ACCEPTED REQUESTS
LOG DONATION DATE
VIEW DONATION HISTORY
MANAGE MY REQUESTS
LOGOUT
You are now available
Home
Donors
Requests
Profile

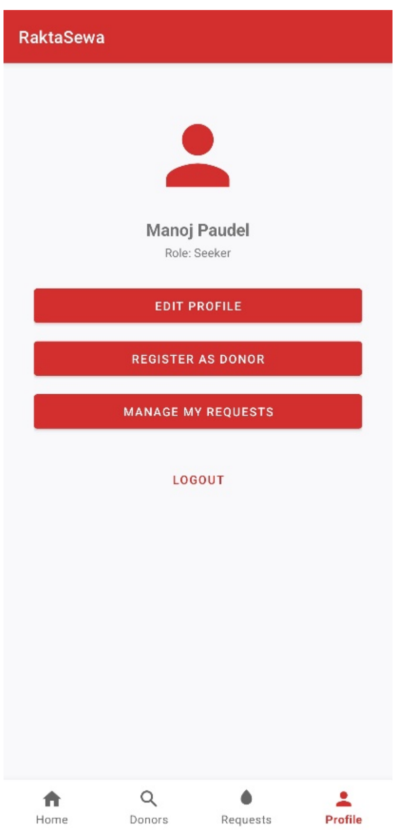

7:59
58%
RaktaSewa
Find Donors
A+
SEARCH DONORS
Found available donors within 5 km.
Master
Available
Blood Type: A+
Phone: 9766608092
Address: pkr
Last Donated: Not available
CONTACT DONOR
Deep
1.1 km
Available
Blood Type: A+
Phone: 9864435090
Address: pokhara
City: Pokhara
Last Donated: Not available
CONTACT DONOR
Home
Donors
Requests
Profile

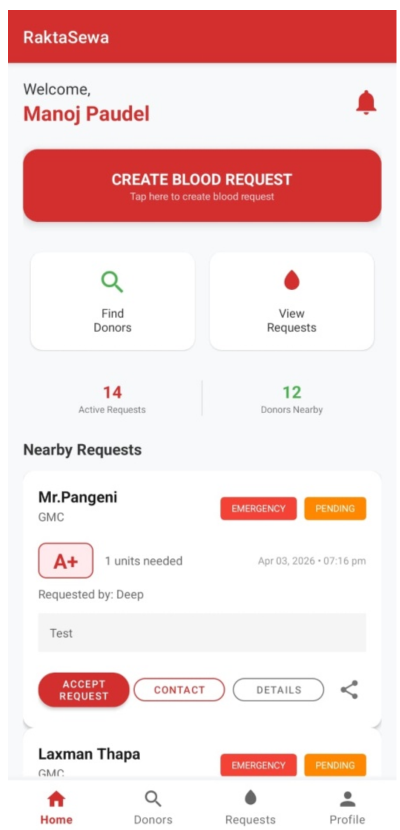

7:58
58%
Blood Requests
Matching your blood group: B+
VIEW ALL
Ram Thapa
PENDING
Manipal
B+
1 units needed
Apr 03, 2026 • 07:58 am
Requested by: Deep
test
ACCEPT REQUEST
CONTAC
DETAILS
+
Home
Donors
Requests
Profile

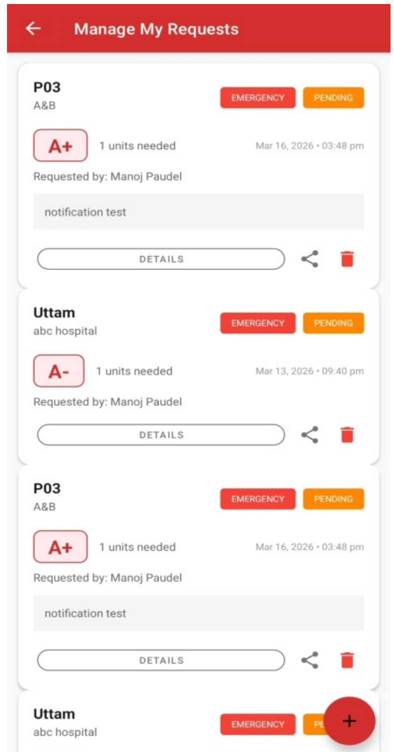

7:59
58%
My Accepted Requests
Ram Thapa
ACCEPTED
Manipal
B+
1 units needed
Apr 03, 2026 • 07:58 am
Requested by: Deep
Accepted by: Pratik Pangeni
test
CONTACT
DETAILS


7:59
58%
RaktaSewa
Pratik Pangeni
cheemsbaltez123@gmail.com
+9779846012133
Blood Type
B+
UPDATE MY LOCATION
Location already saved
SAVE CHANGES


Namaste • Namaste
8:02 Fri, 3 Apr
Blood Needed 8:01 am
A request for A+ blood has been made in your area.
Request Accepted! 7:59 am
Pratik Pangeni has accepted your blood request.
Notification settings
Clear


RaktaSewa
Manoj Paudel
Role: Seeker
EDIT PROFILE
REGISTER AS DONOR
MANAGE MY REQUESTS
LOGOUT
Home
Donors
Requests
Profile


RaktaSewa
Welcome,
Manoj Paudel
CREATE BLOOD REQUEST
Tap here to create blood request
Find Donors
View Requests
14
Active Requests
12
Donors Nearby
Nearby Requests
Mr.Pangeni
GMC
EMERGENCY
PENDING
A+
1 units needed
Apr 03, 2026 • 07:16 pm
Requested by: Deep
Test
ACCEPT REQUEST
CONTACT
DETAILS
Laxman Thapa
EMERGENCY
PENDING
Home
Donors
Requests
Profile


Manage My Requests
P03
A&B
EMERGENCY
PENDING
A+
1 units needed
Mar 16, 2026 • 03:48 pm
Requested by: Manoj Paudel
notification test
DETAILS
Uttam
abc hospital
EMERGENCY
PENDING
A-
1 units needed
Mar 13, 2026 • 09:40 pm
Requested by: Manoj Paudel
DETAILS
P03
A&B
EMERGENCY
PENDING
A+
1 units needed
Mar 16, 2026 • 03:48 pm
Requested by: Manoj Paudel
notification test
DETAILS
Uttam
abc hospital
EMERGENCY

Reference

[1] S. Khan, "Banke faces blood shortage with only 150 pints left in stock," Rising Nepal Daily, 15 June 2024. [Online]. Available: https://risingnepaldaily.com/news/44447.

[2] "Jhapa Red Cross appeals for blood donation amid acute shortage," Kharhub.com, 27 October 2025. [Online]. Available: https://english.khabarhub.com/2025/27/503083/.

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[4] "Where's the blood?," Nagarik Network, 23 November 2012. [Online]. Available: https://myrepublica.nagariknetwork.com/news/53901/.